

① The first statement is true & second is false.

(a) Since $\mathcal{F}$ is a σ -field, we have that

$$S \in \mathcal{F}.$$

Since $\bar{h}'(S) = \Omega$, it follows that $\Omega \in \bar{h}'(\mathcal{F})$.

Next, let $A \in \bar{h}'(\mathcal{F})$ be arbitrary.

This implies $\exists B \in \mathcal{F}$ s.t.

$$A = \bar{h}'(B).$$

$$\text{Now, } A^c = (\bar{h}'(B))^c$$

$$= \bar{h}'(B^c).$$

Since $\mathcal{F}$ is a σ -field, we have $B^c \in \mathcal{F}$.

This implies that $A^c \in \bar{h}'(\mathcal{F})$, which shows that $\bar{h}'(\mathcal{F})$ is closed under the complement operation.

Next suppose the sequence (A_n) be such that

$$A_n \in \tilde{h}'(\mathcal{G}).$$

This implies, for each n , $\exists B_n \in \mathcal{G}$ such that

$$A_n = \tilde{h}^{-1}(B_n).$$

Furthermore, since

$$\bigcup_{n=1}^{\infty} B_n \in \mathcal{G}$$

$$\text{and } \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} \tilde{h}^{-1}(B_n)$$

$$= \tilde{h}^{-1}\left(\bigcup_{n=1}^{\infty} B_n\right),$$

we have that $\tilde{h}'(\mathcal{G})$ is closed under countable union.

Based on the above arguments, it follows that $\tilde{h}'(\mathcal{G})$ is a σ -field, as desired.

(b) Let $\Omega = \{1, 2\}$

$\mathcal{F} = \{a, b\}$.

Suppose $\mathcal{A} = 2^\Omega$ and

$$h: \Omega \mapsto S$$

is given by

$$h(1) = h(2) = a.$$

Claim: The collection
 $\mathcal{H}(\mathcal{A}) = \{h(A) : A \in \mathcal{A}\},$
 where $h(A) = \{h(x) : x \in A\},$
 is not a σ -field.

Proof: Clearly, for any

$$\emptyset \neq A \in 2^\Omega,$$

$$h(A) = \{a\}.$$

On the other hand, $h(\emptyset) = \emptyset.$

This implies $h(\mathcal{A}) = \{\emptyset, \{a\}\}.$

Since $S \notin h(\mathcal{A})$, we get
 that $h(\mathcal{A})$ is Not a
 σ -field, as desired.

(2)(a). We first show that

$$\mathbb{E} X 1_A = 0$$

for a simple random variable.

For any simple function,

$$X = \sum_{i=1}^m a_i 1_{A_i}$$

for some $a_1, \dots, a_m \in \mathbb{R}$
and $A_1, \dots, A_m$ such that

$$\bigcup_{i=1}^m A_i = \Omega$$

and $A_i \cap A_j = \emptyset$ for $i \neq j$.

Therefore,

$$X 1_A = \sum_{i=1}^m a_i 1_{A_i} 1_A$$

$$= \sum_{i=1}^m a_i 1_{A_i \cap A}.$$

Hence, by definition,

$$\mathbb{E} X 1_A = \sum_{i=1}^m a_i P(A_i \cap A)$$

Since $P(A) = 0$, we have

$$P(A \cap A_i) = 0$$

for all $i = 1, 2, \dots, m$.

This shows that

$$\mathbb{E}X \mathbf{1}_A = 0,$$

as desired.

Next suppose X is a non-negative random variable.

Then, there exist a sequence (X_n) of simple random variables such that

$$0 \leq X_n \uparrow X.$$

Now, $X \mathbf{1}_A$ is also non-negative.

Furthermore,

$$0 \leq X_n \mathbf{1}_A \uparrow X \mathbf{1}_A.$$

and $(X_n 1_A)$ is a sequence of simple random variables.

This implies that

$$\mathbb{E} X 1_A = \lim_{n \rightarrow \infty} \mathbb{E} X_n 1_A$$

$$= \lim_{n \rightarrow \infty} 0 = 0,$$

as desired.

Next, suppose X is a general random variable.

$$\text{Then, } (X 1_A)^+ = \max \{X 1_A, 0\}$$

$$= \left(\max \{X, 0\} \right) 1_A$$

$$= X^+ 1_A.$$

$$\text{Similarly, } (X 1_A)^- = X^- 1_A.$$

$$\text{Now, } \mathbb{E} X^+ 1_A = \mathbb{E} X^- 1_A = 0.$$

$$\text{Hence, } \mathbb{E} X 1_A$$

$$= \mathbb{E} (X 1_A)^+ - \mathbb{E} (X 1_A)^- = 0,$$

as desired.

③ Now, $P(X_n \neq 0) = P(X_n = \frac{1}{n})$

$$= \frac{1}{n^2}.$$

(b) The above fact implies that

$$\sum_{n=1}^{+\infty} P\{X_n \neq 0\} = \sum_{n=1}^{+\infty} \frac{1}{n^2}$$

$$= \frac{\pi^2}{6} < +\infty.$$

(c) Borel-Cantelli lemma then states that

$$P(X_n \neq 0 \text{ i.o.}) = 0.$$

That is,

$$P\left(\bigcap_{n=1}^{+\infty} \bigcup_{k \geq n} \{X_k \neq 0\}\right) = 0$$

Equivalently,

$$P\left(\bigcup_{n=1}^{+\infty} \bigcap_{k \geq n} \{X_k = 0\}\right) = 1.$$

For every $\omega \in \bigcup_{n=1}^{+\infty} \bigcap_{k \geq n} \{X_k = 0\}$,

$$\exists N_0: X_k(\omega) = 0 \quad \forall n \geq N_0.$$

Hence, for all such ω ,

$$\sum_{n=1}^{+\infty} X_n(\omega) = \sum_{k=1}^{N_0-1} X_k(\omega)$$

$$\leq \sum_{k=1}^{N_0-1} \frac{1}{k} < +\infty$$

Since $\sum_{k=1}^n X_k(\omega)$ is

monotonically increasing with n

and $\sum_{k=1}^n X_k(\omega) \leq \sum_{k=1}^{N_0-1} \frac{1}{k}$,

it follows that

$$\sum_{k=1}^n X_k(\omega) \nearrow \sum_{k=1}^{+\infty} X_k(\omega) < +\infty$$

Finally, because

$$\bigcup_{n=1}^{+\infty} \bigcap_{k \geq n} \{X_k = 0\} \text{ has}$$

probability 1, the

desired result follows.

④ For any ω ,

$$\sup_{k \geq n} \frac{X_k(\omega)}{k} \text{ is a}$$

decreasing sequence in n .

$$\text{Hence, } \limsup \frac{X_n(\omega)}{n}$$

$$= \inf_{n \geq 1} \left(\sup_{k \geq n} \frac{X_k(\omega)}{k} \right)$$

$$= \inf_{n \geq N} \left(\sup_{k \geq n} \frac{X_k(\omega)}{k} \right).$$

For any $k \geq 0$, it follows that

$$\left\{ \limsup \frac{X_n}{n} = 0 \right\} \text{ Why?}$$

$$\in T_k = \sigma(X_{k+1}, \dots).$$

In turn, it follows that

$$\left\{ \limsup \frac{X_n}{n} = 0 \right\}$$

$$\in \bigcap_{k=0}^{\infty} T_k = T.$$

The desired result follows from Kolmogorov's 0-1 Law.